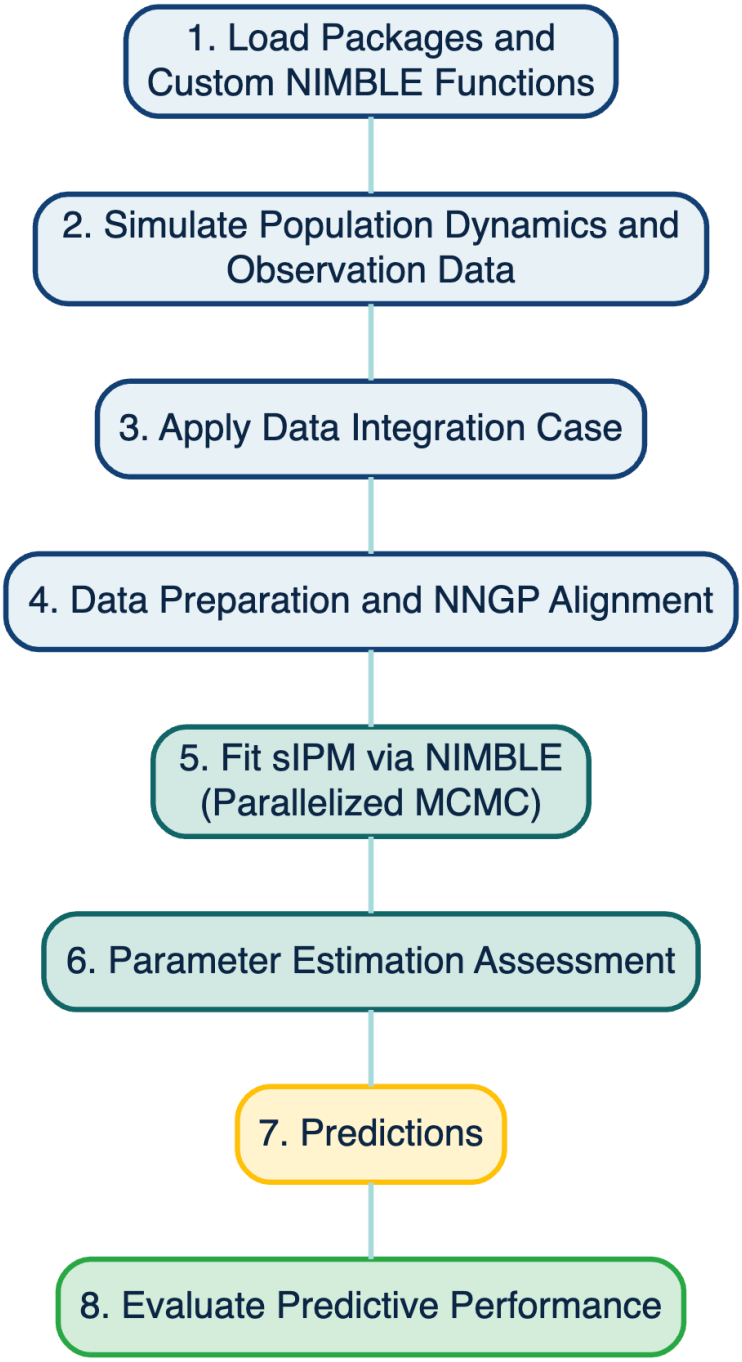
Simulation workflow



The simulation workflow above consisted of eight sequential steps. First, all required R packages and custom NIMBLE functions were loaded and compiled. Second, using the parameter settings described in Section S1.2, latent population dynamics and observation data were simulated. Third, the data integration cases under consideration were applied to the simulated datasets. Fourth, datasets were processed and formatted for model fitting, including construction of the nearest-neighbor structure required for the Nearest-Neighbor Gaussian process (NNGP) approximation.

Fifth, the spatial integrated population model (sIPM) was fitted to each simulated dataset using Markov chain Monte Carlo (MCMC) sampling implemented in NIMBLE, with computation parallelized across simulations. Sixth, parameter estimation performance was evaluated by using the evaluation metrics described in Section \ref{sim_metrics}. Seventh, fitted models were used to generate predictions of population dynamics. Finally, predictive performance was assessed using the prediction metrics described in Section S1.1.